

MASTER DOCUMENT TREE

Helicopter Host Variant – Full Vehicle Package

250 Engineering Documents

Ultra-Safe Central Capsule as Central Survival Core (5.80 m × 3.80 m | 22 t | 7 Occupants)

Aligned to Document 00 Version 3.0 — Comprehensive Engineering Baseline

Document Version	Date	Status	Total Documents
1.0	September 2026	Controlled Tree aligned to Document 00 Version 3.0	250

This Master Document Tree defines the complete set of 250 engineering documents required to design a full helicopter that uses the Ultra-Safe Central Capsule (Large Configuration) as its central survival core. The capsule remains the pure survival module defined in Document 00 Version 1.6 (pure survival module; performance claims remain verification-gated). The helicopter airframe, rotor system, propulsion, flight controls, landing gear, systems and separation interfaces are designed around the capsule. Separation in flight is performed by multi-path release followed by parachute recovery of the capsule. The package covers the complete vehicle from requirements through architecture, structures, aerodynamics, rotor systems, propulsion, avionics, safety, testing and certification considerations.

1. Document Package Overview by Domain

Group	Domain	Doc Range	Count
H1	Program Control, Requirements & Safety Case	001 – 030	30
H2	Vehicle Architecture & Systems Engineering	031 – 060	30
H3	Airframe Structures & Capsule Integration	061 – 100	40
H4	Rotor System, Propulsion & Flight Performance	101 – 140	40
H5	Separation, Safety Layers & Crashworthiness	141 – 175	35
H6	Avionics, Electrical, Flight Controls & Mission Systems	176 – 205	30
H7	Landing Gear, Hydraulics & Mechanical Systems	206 – 225	20
H8	Human Factors, Interiors & Crew Systems	226 – 240	15
H9	Verification, Test, Certification & Support	241 – 250	10
	TOTAL	001 – 250	250

2. Complete Document Tree (001 – 250) with Document Types

All 250 documents are listed below with number, full title and Document Type. The capsule remains the pure survival module (Document 00 Version 1.6). The helicopter is designed around it as the central core.

No.	Document Title	Document Type
GROUP H1 – Program Control, Requirements & Safety Case (001 – 030)		
1	Helicopter Host Master Document Index & Tree (this document)	Management
2	Helicopter Project Charter & Vision	Management
3	Scope Definition – Helicopter with Capsule Core	Requirements
4	Stakeholder Needs & Operational Requirements	Requirements
5	Concept of Operations (ConOps) – Helicopter + Capsule	Operations
6	Design Reference Missions & Mission Profiles	Requirements
7	System Requirements Specification (SRS) – Top Level	Requirements
8	Functional Requirements Specification – Vehicle	Requirements
9	Performance Requirements Specification	Requirements
10	Safety Requirements Specification – Helicopter Host	Safety
11	Environmental Requirements Specification	Requirements
12	Human Factors Requirements – Flight Crew	Requirements
13	Reliability, Availability, Maintainability Requirements	Requirements
14	Interface Requirements – Capsule to Helicopter	Requirements
15	Derived Requirements & Design Constraints	Requirements
16	Requirements Traceability Matrix – Helicopter Package	Management
17	Preliminary Hazard Analysis (PHA)	Safety
18	Functional Hazard Assessment (FHA)	Safety
19	System Safety Assessment Plan	Safety
20	Common Cause Analysis (CCA)	Safety
21	Aircraft-Level FMECA	Safety
22	Fault Tree Analysis – Critical Functions	Safety
23	Safety Case Part 1 – Argument Structure	Safety
24	Safety Case Part 2 – Evidence Framework	Safety
25	Airworthiness Certification Basis Mapping	Certification
26	Means of Compliance Matrix	Certification
27	Particular Risk Analysis (Impact, Fire, Rotor Burst)	Safety
28	Configuration Management Plan	Management
29	Risk Management Plan & Initial Risk Register	Management
30	Program Schedule, Milestones & Resource Outline	Management
GROUP H2 – Vehicle Architecture & Systems Engineering (031 – 060)		
31	Helicopter System Architecture Description	Architecture
32	Functional Architecture – Vehicle Level	Architecture
33	Physical Architecture & Product Breakdown Structure	Architecture
34	Logical Architecture & Data Flow	Architecture
35	Mode & State Definition Document	Architecture
36	Operational Mode Transition Matrix	Architecture
37	Capsule-to-Helicopter Interface Control Document (Master)	ICD
38	Internal Vehicle Interface Control Document	ICD
39	Electrical Power Architecture – Helicopter	Architecture
40	Hydraulic System Architecture	Architecture

No.	Document Title	Document Type
41	Data Architecture & Avionics Network	Architecture
42	Redundancy & Fail-Operational Architecture	Architecture
43	Six-Layer Safety Architecture Integration with Helicopter	Architecture
44	Separation Sequence Logic & Timing (In-Flight)	Design
45	Flight Control Architecture	Architecture
46	Autonomy & Flight Management Architecture	Architecture
47	Health Monitoring & Prognostics Architecture	Architecture
48	Vehicle Mass Properties Control Plan	Management
49	Centre of Gravity & Inertia Management – Complete Vehicle	Analysis
50	Thermal Architecture – Helicopter	Architecture
51	Electromagnetic Compatibility Architecture	Architecture
52	Cyber-Physical Security Architecture	Architecture
53	Model-Based Systems Engineering Model Description	Architecture
54	Trade Study Log – Configuration Decisions	Analysis
55	System Performance Budgets (Mass, Power, Volume, Timing)	Analysis
56	System Integration Strategy	Management
57	End-to-End Mission Timeline – Helicopter Operations	Operations
58	Open Issues & Technical Debt Register	Management
59	Design Review Process & Checklist Library	Management
60	Systems Engineering Management Plan – Helicopter	Management
GROUP H3 – Airframe Structures & Capsule Integration (061 – 100)		
61	Airframe Structural Design Philosophy & Criteria	Requirements
62	Primary Airframe Structure Design Description	Design
63	Capsule Bay Structure Design	Design
64	Capsule Attachment Hard-Points Design (Shock-Absorbing)	Design
65	Load Path Analysis – Capsule to Airframe	Analysis
66	Fuselage Structure Design	Design
67	Tail Boom & Empennage Structure Design	Design
68	Landing Gear Attachment Structure Design	Design
69	Engine / Transmission Mount Structure Design	Design
70	Rotor Pylon Structure Design	Design
71	Door, Hatch & Access Panel Structural Design	Design
72	Material Selection Report – Airframe Composites & Metals	Materials
73	Material Allowables & Design Values – Airframe	Materials
74	Joint Design & Fastener Specification – Airframe	Specification
75	Corrosion Protection & Coatings – Airframe	Specification
76	Airframe Mass Breakdown & Optimization	Analysis
77	Finite Element Model Description – Global Airframe	Analysis
78	Finite Element Model – Capsule Integration Region	Analysis
79	Static Load Analysis Report – Airframe	Analysis
80	Dynamic & Modal Analysis Report – Airframe	Analysis
81	Crash & Impact Analysis – Airframe + Capsule	Analysis

No.	Document Title	Document Type
82	Fatigue & Damage Tolerance Analysis – Airframe	Analysis
83	Buckling & Stability Analysis	Analysis
84	Bird Strike & Foreign Object Damage Assessment	Analysis
85	Structural Safety Factor Justification – Airframe	Analysis
86	Damage Tolerance & Fail-Safe Assessment – Airframe	Safety
87	Residual Strength after Capsule Separation	Analysis
88	Structural Test Requirements Specification – Airframe	Test
89	Structural Test Plan – Static	Test
90	Structural Test Plan – Dynamic / Fatigue	Test
91	Capsule-to-Airframe Interface Strength Test Plan	Test
92	Critical Structural Items List – Helicopter	Management
93	Structural Design Review Checklist – Airframe	Review
94	Airframe Structural Design Summary	Management
95	Weight & Balance Report – Complete Helicopter	Analysis
96	Inertia & Mass Distribution Report	Analysis
97	Structural Interface Control Document – Capsule Bay	ICD
98	Manufacturing Considerations – Airframe	Manufacturing
99	Assembly Sequence – Capsule into Airframe	Manufacturing
100	Airframe Family Document Index	Management
GROUP H4 – Rotor System, Propulsion & Flight Performance (101 – 140)		
101	Rotor System Requirements Specification	Requirements
102	Main Rotor System Architecture	Architecture
103	Main Rotor Blade Design Description	Design
104	Main Rotor Hub & Control System Design	Design
105	Tail Rotor / Anti-Torque System Design	Design
106	Rotor System Dynamics & Stability Analysis	Analysis
107	Rotor Loads Analysis	Analysis
108	Rotor Performance Analysis	Analysis
109	Rotor System Mass Properties	Analysis
110	Rotor System FMECA	Safety
111	Propulsion System Requirements	Requirements
112	Engine Selection & Installation Design	Design
113	Transmission / Gearbox Design Description	Design
114	Drive Shaft & Coupling Design	Design
115	Engine Mount & Vibration Isolation Design	Design
116	Fuel System Design Description	Design
117	Propulsion System Performance Analysis	Analysis
118	Propulsion System FMECA	Safety
119	Flight Performance Requirements	Requirements
120	Performance Analysis – Hover, Climb, Cruise, Range	Analysis
121	Weight & Performance Sensitivity Study	Analysis
122	Flight Envelope Definition	Requirements

No.	Document Title	Document Type
123	Handling Qualities Requirements	Requirements
124	Stability & Control Analysis	Analysis
125	Autorotation Performance Analysis	Analysis
126	One-Engine-Inoperative Performance	Analysis
127	Noise Requirements & Assessment	Requirements
128	Vibration Requirements & Assessment	Requirements
129	Rotor & Propulsion Interface Control Document	ICD
130	Rotor System Test Plan Outline	Test
131	Propulsion System Test Plan Outline	Test
132	Performance Flight Test Requirements	Test
133	Rotor System Critical Design Review Checklist	Review
134	Propulsion Critical Design Review Checklist	Review
135	Performance Validation Plan	Test
136	Rotor Blade Bird Strike & Lightning Considerations	Safety
137	Transmission Chip Detection & Health Monitoring	Design
138	Engine Control System Interface Requirements	Requirements
139	Propulsion & Rotor Family Risk Assessment	Safety
140	Rotor & Propulsion Family Document Index	Management
GROUP H5 – Separation, Safety Layers & Crashworthiness (141 – 175)		
141	In-Flight Separation System Requirements	Requirements
142	Separation Architecture – Helicopter Host	Architecture
143	Pyrotechnic Separation Path Design – Airframe	Design
144	Hydraulic / Mechanical Separation Path Design	Design
145	Pure Mechanical Manual Separation Design	Design
146	Separation Sequence Timing & Interlocks – Flight	Design
147	Clean-Break Disconnect Design – All Interfaces	Design
148	Separation Dynamics Analysis – In-Flight Release	Analysis
149	Re-contact & Rotor Interference Risk Assessment	Safety
150	Separation System FMECA – Helicopter	Safety
151	Crew-Commanded vs Automatic Separation Criteria	Requirements
152	Integration of Capsule Six Safety Layers with Helicopter	Architecture
153	Layer 1–4 Activation Logic in Helicopter Context	Design
154	Post-Separation Capsule Trajectory & Clearance Analysis	Analysis
155	Crashworthiness Requirements – Complete Vehicle	Requirements
156	Crash Load Cases – Helicopter + Capsule	Safety
157	Energy Absorption Strategy – Airframe & Landing Gear	Design
158	Occupant Protection under Helicopter Crash Loads	Analysis
159	Fire Protection System – Helicopter Airframe	Design
160	Fire Detection & Suppression Logic – Vehicle Level	Design
161	Emergency Flotation System – Helicopter (if required)	Design
162	Ditching Performance & Capsule Release after Ditching	Analysis
163	Safety Assessment – Combined Helicopter + Capsule	Safety

No.	Document Title	Document Type
164	Particular Risks – Rotor Burst, Engine Fire, Fuel Leak	Safety
165	Separation System Interface Control Document	ICD
166	Crashworthiness Test Plan Outline	Test
167	Separation System Test Plan – Ground & Flight	Test
168	Safety Critical Design Review Checklist	Review
169	Airworthiness Safety Case Contribution – Separation	Certification
170	Emergency Procedures – Capsule Separation in Flight	Operations
171	Post-Separation Recovery Coordination Procedures	Operations
172	Safety Family Risk Register – Helicopter Host	Safety
173	Independence Demonstration Plan – Separation Paths	Safety
174	Certification Considerations – Separation System	Certification
175	Separation & Safety Family Document Index	Management
GROUP H6 – Avionics, Electrical, Flight Controls & Mission Systems (176 – 205)		
176	Avionics Architecture Description – Helicopter	Architecture
177	Electrical Power System Design – Vehicle Level	Design
178	Emergency Power & Battery Bus Integration with Capsule	Design
179	Flight Control System Architecture	Architecture
180	Flight Control Computer Specification	Specification
181	Primary Flight Control Laws & Modes	Design
182	Automatic Flight Control System Requirements	Requirements
183	Sensor Suite Requirements – Helicopter	Requirements
184	Navigation System Requirements	Requirements
185	Communication System Requirements	Requirements
186	Display & Crew Alerting System Requirements	Requirements
187	Human-Machine Interface Design – Cockpit	Design
188	Health Monitoring System Design	Design
189	Mission System Architecture (if applicable)	Architecture
190	Data Bus & Network Architecture	Architecture
191	Software Architecture & Partitioning – Vehicle	Software
192	Software Development Assurance Levels	Software
193	Software Requirements – Safety Critical Functions	Software
194	Electromagnetic Compatibility Requirements – Vehicle	Requirements
195	Lightning Protection Design	Design
196	Cybersecurity Requirements – Flight Critical Systems	Security
197	Avionics Environmental Qualification Requirements	Requirements
198	Avionics Interface Control Document	ICD
199	Avionics Mass & Power Budget	Analysis
200	Avionics Test & Verification Plan Outline	Test
201	Flight Control System Test Plan Outline	Test
202	Avionics Family Risk Assessment	Safety
203	Avionics Critical Design Review Checklist	Review
204	Cockpit Design Review Checklist	Review

No.	Document Title	Document Type
205	Avionics & Flight Control Family Document Index	Management
	GROUP H7 – Landing Gear, Hydraulics & Mechanical Systems (206 – 225)	
206	Landing Gear Requirements Specification	Requirements
207	Landing Gear Architecture & Configuration	Architecture
208	Main Landing Gear Design Description	Design
209	Nose / Tail Landing Gear Design Description	Design
210	Landing Gear Shock Absorption Analysis	Analysis
211	Landing Gear Loads & Strength Analysis	Analysis
212	Landing Gear Retraction / Extension System Design	Design
213	Hydraulic System Design Description	Design
214	Hydraulic Power Generation & Distribution	Design
215	Flight Control Hydraulic Actuation Design	Design
216	Utility Hydraulic Systems Design	Design
217	Mechanical Control System Design (if applicable)	Design
218	Landing Gear & Hydraulics FMECA	Safety
219	Landing Gear Interface Control Document	ICD
220	Landing Gear Test Plan Outline	Test
221	Hydraulic System Test Requirements	Test
222	Landing Gear Critical Design Review Checklist	Review
223	Ground Handling & Towing Provisions	Design
224	Landing Gear Family Risk Assessment	Safety
225	Landing Gear & Hydraulics Family Document Index	Management
	GROUP H8 – Human Factors, Interiors & Crew Systems (226 – 240)	
226	Human Factors Engineering Requirements – Helicopter	Requirements
227	Cockpit Layout & Ergonomics Design	Design
228	Pilot & Co-Pilot Station Design	Design
229	Crew Seat Design & Energy Absorption	Design
230	Restraint System Design – Flight Crew	Design
231	Emergency Egress Design – Helicopter Airframe	Design
232	Egress Path Analysis after Capsule Separation	Analysis
233	Internal Lighting, Marking & Wayfinding	Design
234	Noise, Vibration & Cabin Comfort Requirements	Requirements
235	Crew Workload Assessment Methodology	Analysis
236	Human Factors Test & Evaluation Plan Outline	Test
237	Occupant Injury Criteria – Helicopter Crash	Safety
238	Crew Training & Emergency Procedure Outline	Operations
239	Human Factors Critical Design Review Checklist	Review
240	Human Factors Family Document Index	Management
	GROUP H9 – Verification, Test, Certification & Support (241 – 250)	
241	Helicopter Verification & Validation Master Plan	Test
242	Ground Test Plan Outline – Complete Vehicle	Test
243	Flight Test Plan Outline	Test

No.	Document Title	Document Type
244	Integrated Capsule Separation Flight Test Requirements	Test
245	Certification Program Plan – Helicopter Host	Certification
246	Type Certification Data Sheet Outline	Certification
247	Continued Airworthiness & Maintenance Concept	Support
248	Spare Parts & Support Equipment Requirements	Support
249	Helicopter Host Risk Register – Master	Management
250	Helicopter Host Package Document Index & Traceability	Management

This 250-document Helicopter Host Variant Package is controlled against Document 00 Version 1.6. The Ultra-Safe Central Capsule remains the pure survival module (no under-surface rotors). The helicopter airframe, rotor system, propulsion and all vehicle systems are designed around the capsule as the central survival core. In-flight separation is multi-path followed by parachute recovery of the capsule. All numerical values, safety principles and interface rules shall remain consistent with the locked baseline of the capsule.

— End of Helicopter Host Variant Master Document Tree (250 Documents) —
Ultra-Safe Central Capsule | Document 00 Version 1.6 | Helicopter Host Package
© 2026 – Confidential – Founder: Amirbahador Atai